

1. recall that the Earth's lithosphere (the rigid outer layer of Earth made up of the crust and the part of the mantle just below it) is made up of a mixture of minerals;
 2. recall that silicon, oxygen and aluminium are very abundant elements in the crust;
 3. be able to interpret data about the abundances of elements in rocks.
 4. recall that much of the silicon and oxygen is present in the Earth's crust as the compound silicon dioxide;
 5. recall the properties of silicon dioxide (for example hardness, melting point, electrical conductivity and solubility in water);
 6. explain the properties of silicon dioxide in terms of a giant structure of atoms held together by strong covalent bonding (for example melting point, boiling point, hardness, solubility and electrical conductivity)
 7. understand that silicon dioxide is found as quartz in granite, and is the main constituent of sandstone;
 8. understand that some minerals are valuable gemstones because of their rarity, hardness and appearance;
 9. **interpret data and explain the uses and properties of other elements and compounds with giant covalent structures (no recall expected).**
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